

MANI YADAV

DATA SCIENCE | PYTHON • SQL • REMOTE SENSING • GOOGLE EARTH ENGINE

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PROFESSIONAL SUMMARY

Computer Science undergraduate focused on Python-based data analysis, environmental datasets, remote sensing, and Earth observation. Hands-on experience with data cleaning, exploratory analysis, visualization, satellite imagery, NDVI, SAR change detection, and Google Earth Engine. Interested in climate data science, climate modeling, geospatial analytics, and using quantitative methods to understand environmental change.

TECHNICAL SKILLS

Data Science: Python, Pandas, NumPy, Scikit-learn, Exploratory Data Analysis (EDA), Data Cleaning, Data Validation, Statistics & Probability **Visualization:** Matplotlib, Plotly, Excel, Google Sheets

Geospatial & Remote Sensing: Google Earth Engine, QGIS, Rasterio, GeoPandas, Folium, NDVI, SAR, Satellite Imagery

Databases & Tools: SQL, Git, GitHub, Jupyter Notebook

PROJECTS

Vegetation Change Analysis using NDVI | Python, QGIS, Rasterio | [GitHub](#)

- Processed satellite imagery from 2022 and 2026 to calculate NDVI, perform raster analysis, spatial masking, and change detection across a 523.80 km² study area.
- Quantified 145.94 km² vegetation gain, 20.32 km² loss, and 357.54 km² stable vegetation using a ± 0.05 NDVI-change threshold.

Fitness Data Cleaning & Exploratory Data Analysis | Python, Pandas, Matplotlib | [GitHub](#)

- Audited a real-world fitness dataset using Pandas to identify missing values, inconsistent date formats, duplicate rows, and data-entry anomalies.
- Standardized dates, median-imputed missing calorie values, corrected a 450-minute workout entry to 45 minutes, and removed duplicate records.
- Created visual analyses of workout frequency, pulse relationships, and calorie expenditure; identified a strong linear relationship between average resting pulse and peak pulse.

Online Retail Sales Analysis | Python, Pandas, Matplotlib | [GitHub](#)

- Built a Python data pipeline to clean and analyze 541,909 e-commerce transactions, resolving 136K+ missing values and reaching 95% data quality.
 - Analyzed sales patterns and customer geography, identifying revenue opportunities worth £500K+ and a UK-dominant market share of 45%.
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EXPERIENCE

Project Intern — A Rocha India | Jun 2026 – Jul 2026 | Bengaluru, Karnataka

- Collected GPS-based field data and boundary information; digitized, structured, and validated field observations for spatial analysis.
- Organized field observations and mapped elephant movement, habitat characteristics, and human–wildlife conflict data using Google Sheets and mapping workflows.

Summer Intern, Electrical & Instrumentation — Aditya Birla Group | Jul 2026 – Aug 2026 | Manikgarh Cement Works, Maharashtra

- Gained hands-on exposure to sensors, transmitters, motors, control panels, and industrial process-monitoring systems.
 - Studied instrumentation and process flow in coal and grinding mills, building an understanding of monitoring and control systems.
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EDUCATION

Bachelor of Technology (B.Tech) — Computer Science | 2023 – Present

Aravali Institute of Technical Studies, Udaipur (Affiliated to Rajasthan Technical University, Kota)

Relevant Coursework: Digital Image Processing, Machine Learning, Artificial Intelligence, Cloud Computing

CERTIFICATION

Probability and Statistics for Data Science — GUVI